Applied Economics

Multiple Linear Regression Model II

Pello Aspuru

Winter 2025/26

Plan for this Lecture

Last lecture:

- Introduced multiple linear regression model (MLR)
- Defined population parameters β, estimators and estimates ($\hat{\beta}$)
- Derived Ordinary Least Squares (OLS) estimator.

Today lecture:

1. Classical MLR Assumptions MLR.1 - MLR.4
2. OLS Properties: Unbiasedness, Consistency and Asymptotic Normality
3. Nonlinear Functional Forms: Polynomials, Logarithmic Transformations, Group Dummies, Interaction Terms
4. Empirical Examples in R

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MLR Assumptions

Motivating Example

Economic question: does pollution cause a decrease of housing prices in Vienna?

MLR model for the population: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i \quad i = 1, \dots N$

- Outcome variable $Y \in [0, \infty]$: housing prices
- Regressors: pollution concentration $X_{1i} \in (0, \infty)$, house size $X_{2i} \in (0, \infty)$.

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population <- read.csv("./AE_week3_MLR2/df_pop_house_prices.csv")
mlr_params = lm(price ~  nox + rooms, data = population)
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Equivalent model for observed sample n: $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{u}_i \quad i = 1, \dots n$

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n <- 100
df <- population[sample(nrow(population), n), ]
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$\rightarrow$ β is a number representing the population, $\hat{\beta}$ is a random variable.

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1. Linearity in parameters $\rightarrow$ closed-form, tractable OLS estimator function.
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Assumption 1 - Linear Parameters

In **population**, regressors $\mathbf{X}_i$ determine the outcome Y_i through linear parameters β:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i \qquad \text{Linear parameters: Yes}$$

$$Y_i = \beta_0 + X_{1i}^{\beta_1} + \frac{X_{2i}}{\beta_2} + u_i \qquad \text{Non-linear parameters: No}$$

$$Y_i = \beta_0 + \beta_1 ln(X_i) + u_i \qquad \text{Non-linear regressors: ?}$$

Why do we need parameters to be linear?

Linear parameters enable matrix form notation of the sample model, consequent straightforward derivation of $\hat{\beta}$ formula:

$$\mathbf{Y} = \mathbf{X} \times \mathbf{b} + \hat{\mathbf{u}} \quad \rightarrow \quad SSR(b) = \hat{\mathbf{u}}'\hat{\mathbf{u}} \quad \xrightarrow{\text{FOC wrt b}} \quad \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

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Assumption 2: No perfect multicollinearity

Multicolinearity: Regressors $X_1, X_2 ...$ are **highly** correlated.

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multicolinearity = lm(nox ~ rooms, data = df)
summary(multicolinearity)
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Perfect Multicollinearity: a regressor X_1 is a **perfect** linear function of other regressors:

$$X_1 = 1, \qquad X_1 = X_1 + \sigma \cdot X_2 \qquad X_1 = 1 - X_2$$

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df$bedrooms = df$rooms - 1
ols_est_perf_mc = lm(price ~ nox + rooms + bedrooms, data = df)
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With perfect multicolinearity, the matrix $(\mathbf{X}'\mathbf{X})$ is not invertible $\rightarrow$ OLS estimator is not computable.

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Assumption 3: Random Sampling

We observe an **i.i.d.** random sample of the population, comprised of n independent observations:

$$\{Y_i, X_{i1}, X_{i2}, \ldots, X_{ik}\}, \; i = 1, 2, \ldots, n$$

This assumption implies independence across observations i.

$$\mathbf{X}_i \perp \mathbf{X}_{i'} \text{ and } u_i \perp u_{i'} \text{ for all } i' \neq i \quad \Rightarrow \quad E(u_i u_j \mid \mathbf{X}) = 0 \quad \text{for all } i \neq j$$

Why do we need the Random Sampling Assumption?

For the derivation of the OLS estimate $\hat{\beta}$ variance (next session).

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Assumption 4: Exogeneity[1]

Conditional Mean Independence Assumption $\rightarrow$ Exogeneity Assumption

$$E[u_i|X_i = A + 1, X_{2i}, ..., X_{ki}] = E[u_i|X_i = A, X_{2i}, ..., X_{ki}] = 0$$

The error term u_i has conditional expected value of zero for all regressors:

$$E[u_i|X_{i1}, X_{i2}, \cdots, X_{ik}] = 0$$

Why do we need the Exogeneity assumption?

1. As with conditional mean independence assumption: causal interpretation of β.
2. Guarantees estimate $\hat{\beta}$ is a "good" representation of parameter β.

[1] Also called **Zero conditional expectation** assumption

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Unbiasedness of OLS - Theorem

The expectation of parameter estimate $\hat{\beta}_j$ equals the true parameter β_j:

$$E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \ldots, k$$

```
n <- 100
reps <- 1000
beta_1 <- mlr_params$coefficients["nox"] # parameter
beta_hat_vec <- numeric(reps) # Estimates
```

Estimate $\hat{\beta}_1$ *reps* $= 1000$ times, each with a **different** sample of size $n = 100$.

```
for (i in 1:reps) {
    df <- population[sample(nrow(population), n), ]
    fit <- lm(price ~ nox + rooms, data = df)
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Unbiasedness of OLS - Graphical Illustration

Plot all $\hat{\beta}_1$ OLS estimates, together with β_1 true value they aim to represent:

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sims <- data.frame(beta_hat = beta_hat_vec)
ggplot(sims, aes(x = beta_hat)) +
geom_histogram(bins = 40, alpha = 0.8) +
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Unbiasedness - Formal Derivation

$\underset{n\times 1}{\mathbf{Y}}$ can be expressed in terms of population parameters as: $\mathbf{Y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$

Decompose OLS estimator into population parameter vector $\boldsymbol{\beta}$ and a sampling error $\hat{\mathbf{u}}$:

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Next, take the conditional expectations:

$$\begin{aligned} E[\hat{\boldsymbol{\beta}}|\mathbf{X}] &= E[\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}|\mathbf{X}] \\ &= E[\boldsymbol{\beta}|\mathbf{X}] + E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}|\mathbf{X}] \\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\underbrace{E(\mathbf{u}|\mathbf{X})}_{=\mathbf{0}_{(n\times 1)}} \\ &= \boldsymbol{\beta} \qquad \text{Exogeneity} \end{aligned}$$

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Unbiasedness - Formal Derivation

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Consistency of the OLS Estimator - Theorem

The OLS estimator $\hat{\boldsymbol{\beta}}$ converges in probability towards $\boldsymbol{\beta}$:

$$\hat{\beta}_j \xrightarrow{p} \beta_j \text{ for all } j = 0, 1, ..., k$$

Proof sketch

1. Take a parameter estimate random draw $\hat{\beta}_j$, estimated from a sample of size n.
2. Define the gap between the estimate and the parameter: $|\hat{\beta}_j - \beta_j|$
3. Gap is large if $|\hat{\beta}_j - \beta_j| > \epsilon$, for some positive number $\epsilon > 0$.
4. The probability that the gap is large is: $\mathbf{P}(|\hat{\beta}_j - \beta_j| > \epsilon) \in [0, 1]$.
   - Large gap not likely $\rightarrow$ $\mathbf{P}(|\hat{\beta}_j - \beta_j| > \epsilon) \approx 0$

As sample size goes to ∞, the large gap **becomes less and less likely**:

$$\lim_{n \to \infty} \mathbf{P}(|\hat{\beta}_j - \beta_j| > \epsilon) = 0 \text{ for all } \epsilon > 0$$

More in appendix.

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Consistency of the OLS Estimator - Graphical Illustration

Find $|\hat{\beta}_j - \beta_j|$ gap *reps* $= 1000$ times, each with a **different** sample of size $n = 100$.

```
gaps_beta_hat_beta <- numeric(reps)
for (i in 1:reps) {
df <- population[sample(nrow(population), n), ]
fit <- lm(price ~ nox + rooms, data = df)
gap <- abs(coef(fit)["nox"]-beta_1)
gaps_beta_hat_beta[i] <- gap}
```

Plot the gaps:

```
sims_gaps <- data.frame(gaps = gaps_beta_hat_beta)
ggplot(sims_gaps, aes(x = gaps)) +
geom_histogram(bins = 40, alpha = 0.8) +
coord_cartesian(xlim = c(0, 2000))
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→ What happens when we set $n = 10000$?

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The random variable $\hat{\boldsymbol{\beta}}$ is asymptotically normally distributed:

$$\hat{\boldsymbol{\beta}} \approx N\left(\boldsymbol{\beta}, \frac{1}{n}\Sigma\right)$$

Proof sketch:

1. Rewrite gap between estimator and parameter: $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\left(\frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{u}\right)$
2. By law of large numbers + no perfect multicolinearity $\frac{1}{n}\mathbf{X}'\mathbf{X} \xrightarrow{p} \underset{k\times k}{\mathbf{Q}}$
3. By central limit theorem, exogeneity: $\left(\frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{u}\right) \xrightarrow{d} \mathcal{N}(0, \Omega)$
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Recap: OLS Assumptions and Properties

- We revisited the OLS assumptions and the properties we get with them.
- If the assumptions hold, our estimates $\hat{\beta}$ are correct on average (unbiased) and accurate in large samples (consistent).
- Next session:
  - How does the variance matrix $Var(\hat{\beta})$ look like?
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Nonlinear Functional Forms

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OLS Assumption 1 requires linearity only in **parameters**. OLS estimation can incorporate nonlinear transformations of the **variables**.

Why include nonlinear transformations?

- Linearity implies constant partial effects (independent of the level of the explanatory variable). This is often implausible, e.g. in case of decreasing marginal returns.
- Logarithmic transformations allow to estimate (semi-)elasticities.

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(i) Polynomials of a regressor

To model non-linear effects of X_{1i} on Y_i include regressor polynomials: $X_{1i}^2, X_{1i}^3 \ldots$

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{1i}^2 + \hat{\beta}_3 X_{2i} + \hat{u}_i \quad i = 1, \ldots n$$

```
df$nox_sq <- (df$nox)^2
ols_est = lm(price ~ nox + nox_sq + rooms, data = df)
summary(ols_est)
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- Linear model: the effect of increasing pollutant X_{1i} in $+1$ units on Y_i is always β_1.
- Quadratic model: the effect of increasing pollutant X_{1i} in $+1$ units on Y_i is $\beta_1 + 2\beta_2 X_{1i}$. Depends on level of X_{1i}.

$\rightarrow$ What does $\beta_2 < 0$ represent? Increasing or decreasing costs?

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(ii) Indicator Variables

A more flexible alternative to capture nonlinear effects is to include indicator/dummy variables for different ranges of a continuous regressor.

Example: Split the X_{2i} house rooms variable into two categories: small and big.

$$Is_Big_i = \begin{cases} 1 & \text{if } X_{2i} > 5 \text{ rooms,} \\ 0 & \text{otherwise.} \end{cases}$$

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The new specification allows the effect of house size to vary non-linearly, depending on whether the house is classified as "big" or "small".

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 Is_Big_i + \hat{u}_i \quad i = 1, ...n$$

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ols_est_indic = lm(price ~ nox + is_big, data = df)
summary(ols_est_indic)
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- Previous $\hat{\beta}_2$ assumed "adding" more rooms to a house had a linear effect on price.
- New $\hat{\beta}_2$ assumes that being a big house causes a discrete jump in its price $\rightarrow$ Non-linear effect!

$\hat{\beta}_2$ is the average difference in house price Y_i between big and small houses, holding remaining regressors X_{1i} (pollution) fixed.

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$\hat{\beta}_2$ is the average difference in house price Y_i between big and small houses, holding remaining regressors X_{1i} (pollution) fixed.

(ii) Indicator Variables

The new specification allows the effect of house size to vary non-linearly, depending on whether the house is classified as "big" or "small".

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 Is_Big_i + \hat{u}_i \quad i = 1, ...n$$

```
ols_est_indic = lm(price ~ nox + is_big, data = df)
summary(ols_est_indic)
```

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(iii) Interaction Terms

We can further enrich our model by interacting indicator Is_Big_i and linear X_{2i} variables to capture differences in slope parameters across different subgroups.

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 Is_Big_i + \hat{u}_i \quad (1)$$

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 Is_Big_i + \hat{\beta}_4 Is_Big_i \times X_{2i} + \hat{u}_i \quad (2)$$

Model (1) assumes that size X_{2i} has the same effect on all types of houses.
Model (2) allows the effect of size to differ across house size categories:

- for small houses: β_2, because $Is_Big_i \times X_{2i} = 0$
- for big houses: $\beta_2 + \beta_4$, because $Is_Big_i \times X_{2i} = X_{2i}$

```
ols_est_inter = lm(price ~ nox + rooms*is_big, data = df)
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(iv) Logarithmic Transformations

Starting from today's motivating example, turn pollution varaible $X_{1i} \rightarrow ln(X_{1i})$

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 ln(X_{1i}) + \hat{\beta}_2 X_{2i} + \hat{u}_i \quad i = 1, ...n$$

Differentiate the equation with respect of X_{1i} to get the partial effect of X_{1i} on Y_i:

$$\frac{\partial Y_i}{\partial X_{1i}} = \hat{\beta}_1 \frac{\partial ln(X_{1i})}{\partial X_{1i}} = \hat{\beta}_1 \frac{1}{X_{1i}}$$

Multiply both sides by ∂X_{1i}:

$$\partial Y_i = \hat{\beta}_1 \cdot \frac{\partial X_{1i}}{X_{1i}} \quad \xrightarrow{\frac{\partial X_{1i}}{X_{1i}} = 0.01 \equiv 1\% \text{ change}} \quad \partial Y_i = \hat{\beta}_1 \cdot 0.01$$

For a **1%** change in the regressor $\frac{\partial X_{1i}}{X_{1i}} = 0.01$, the change in outcome is $\partial Y_i = \hat{\beta}_1 \cdot 0.01$.

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Level-Log Model: $Y_i = \beta_0 + \beta_1 ln(X_i) + u_i$

```
ols_lev_log = lm(price ~ log_nox + rooms, data = df)
```

- A 1% increase in X_i is associated with a $\frac{\beta_1}{100}$ unit increase in Y_i

Log-Linear Model: $ln(Y_i) = \beta_0 + \beta_1 X_i + u_i$

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- A 1 unit increase in X_i is associated with a $100\beta_1$ % increase in Y_i.

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- This is the elasticity of Y with respect to X!

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Interpreting coefficients with logarithmic transformations

Model	Dep. Var.	Indep. Var.	Interpretation of β
level-level	Y_i	X_i	$\Delta Y_i = \beta \Delta X_i$
level-log	Y_i	$\ln(X_i)$	$\Delta Y_i = \left(\frac{\beta}{100}\right) \%\Delta X_i$
log-level	$\ln(Y_i)$	X_i	$\%\Delta Y_i = (100\beta) \Delta X_i$
log-log	$\ln(Y_i)$	$\ln(X_i)$	$\%\Delta Y_i = \beta \%\Delta X_i$

ΔX and ΔY denote absolute changes (in units).

$\%\Delta X$ and $\%\Delta Y$ denote relative changes (in %).

APPENDIX

Proof Sketch - Consistency of OLS

We can decompose the OLS estimator into the true coefficient vector and a sampling error, and write it as a function of sample means

$$
\begin{aligned}
\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) \\
&= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \\
&= \boldsymbol{\beta} + \left(\sum_{i=1}^{n} \mathbf{X_i}\mathbf{X_i}'\right)^{-1} \sum_{i=1}^{n} \mathbf{X_i} u_i \\
&= \boldsymbol{\beta} + \left(\frac{1}{n}\sum_{i=1}^{n} \mathbf{X_i}\mathbf{X_i}'\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} \mathbf{X_i} u_i
\end{aligned}
$$

Proof Sketch - Consistency of OLS

Next we apply rules for the probability limit, and the Weak Law of Large Numbers.

$$
\begin{aligned}
plim(\hat{\boldsymbol{\beta}}) &= plim(\boldsymbol{\beta}) + plim\left[\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}\mathbf{X_i'}\right)^{-1}\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}u_i\right] \\
&= \boldsymbol{\beta} + plim\left[\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}\mathbf{X_i'}\right)^{-1}\right] plim\left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}u_i\right] \\
&= \boldsymbol{\beta} + \left[plim\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}\mathbf{X_i'}\right)\right]^{-1} plim\left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}u_i\right] \\
&= \boldsymbol{\beta} + \underbrace{E[\mathbf{X_i}\mathbf{X_i'}]^{-1}}_{(k+1)x(k+1)}\underbrace{E(\mathbf{X_i}u_i)}_{\mathbf{0_{(k+1)}}}
\end{aligned}
$$

Recommended Literature

Heiß Chapter 6.1, Chapter 7